

URBAN HEAT MITIGATION

AI / ML ANALYSIS SYSTEM

Geospatial Physics-Informed Analysis & Cooling Optimization

City: Delhi | 2026-08-21 01:33

24.7%

City Hotspot Area

+7.4°C

Urban Heat Excess

1.0000

PIML Model R²

0.47°C

Optimal Cooling

Framework:

Physics-Informed XGBoost + LSTM + SHAP + Getis-Ord Gi*

Data Sources:

Synthetic Landsat LST proxy, ERA5 atmospheric, OSM morphology

Grid:

200×200 cells @ 100m resolution

Interventions:

Urban Greening, Cool Roofs, Green Roofs, Water Bodies, Permeable Pavements, Street Trees

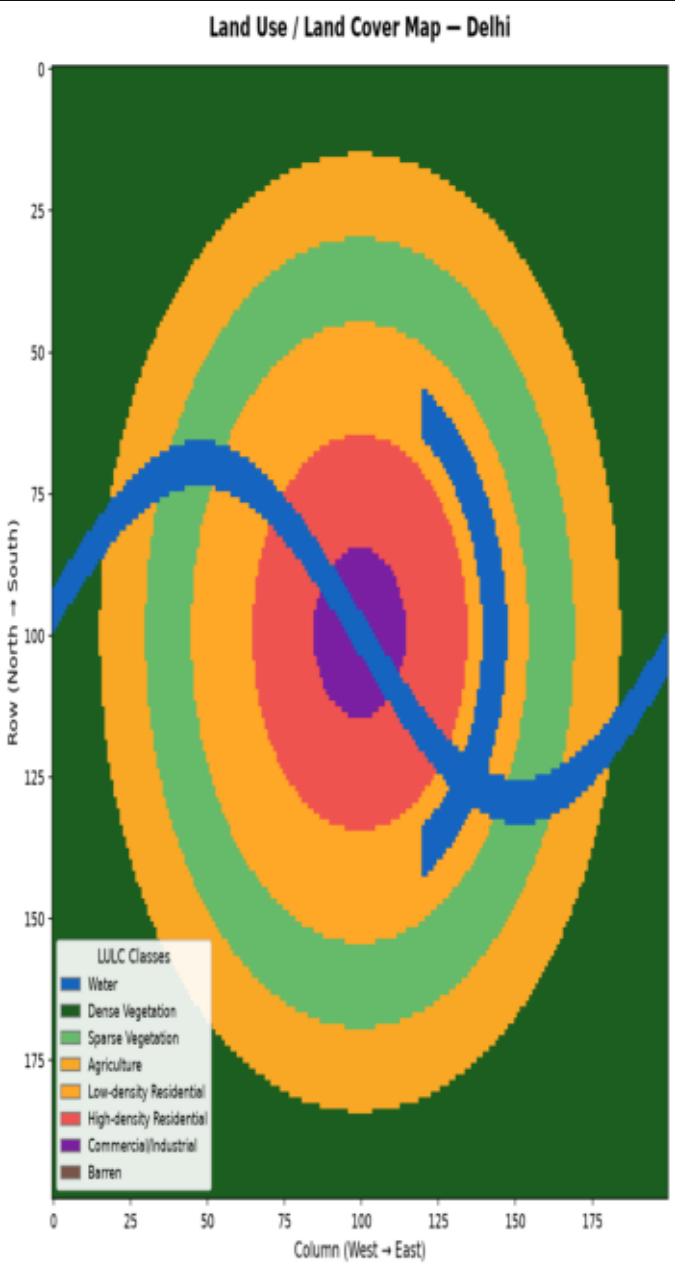
Optimizer:

Greedy budget-constrained, Budget=50,000 units

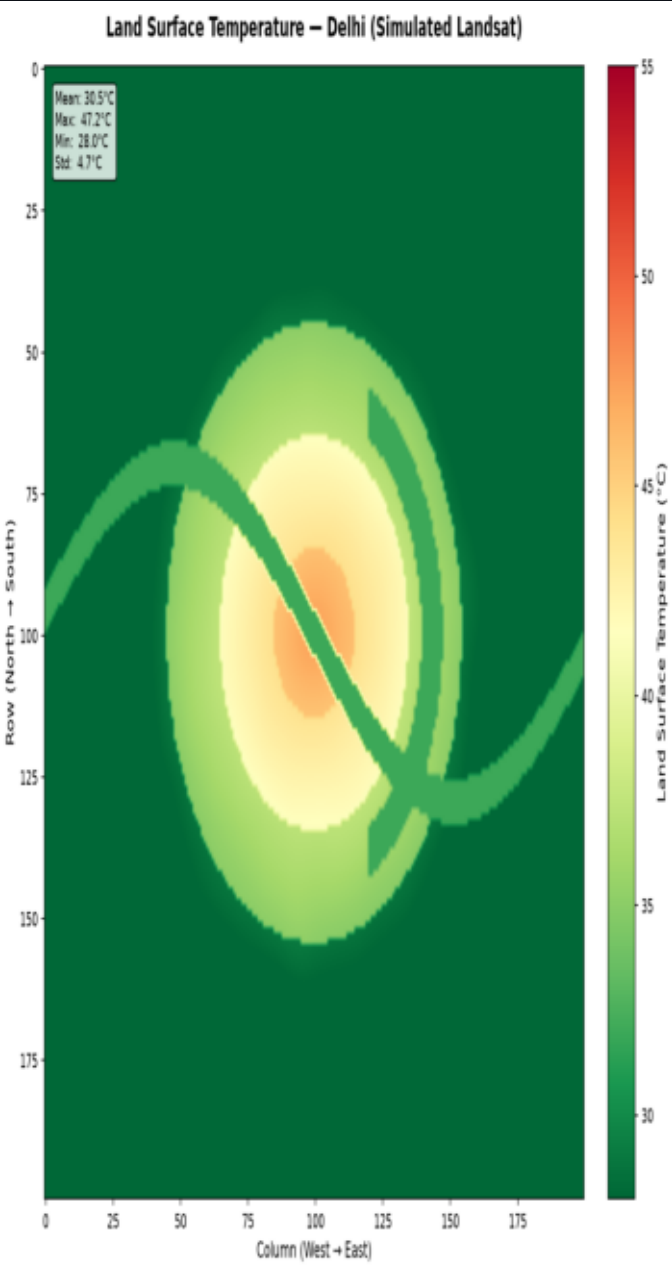
Urban Heat Stress Maps

City: Delhi

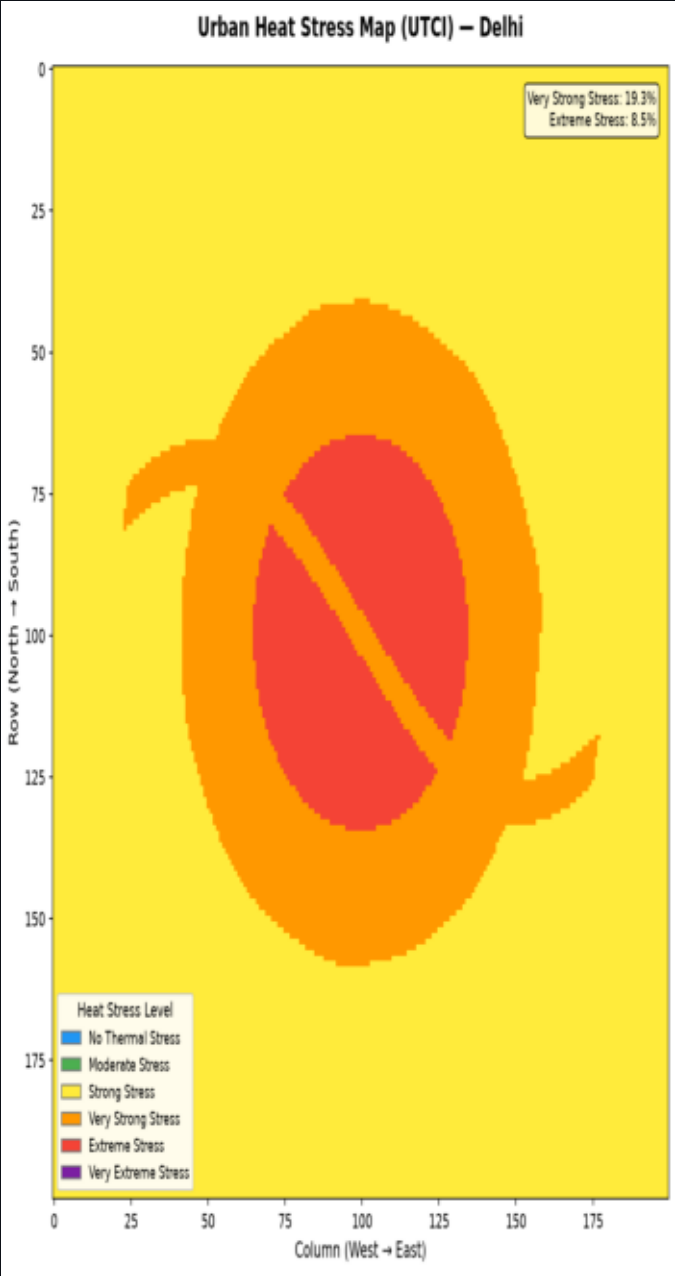
Land Use / Land Cover



Land Surface Temperature



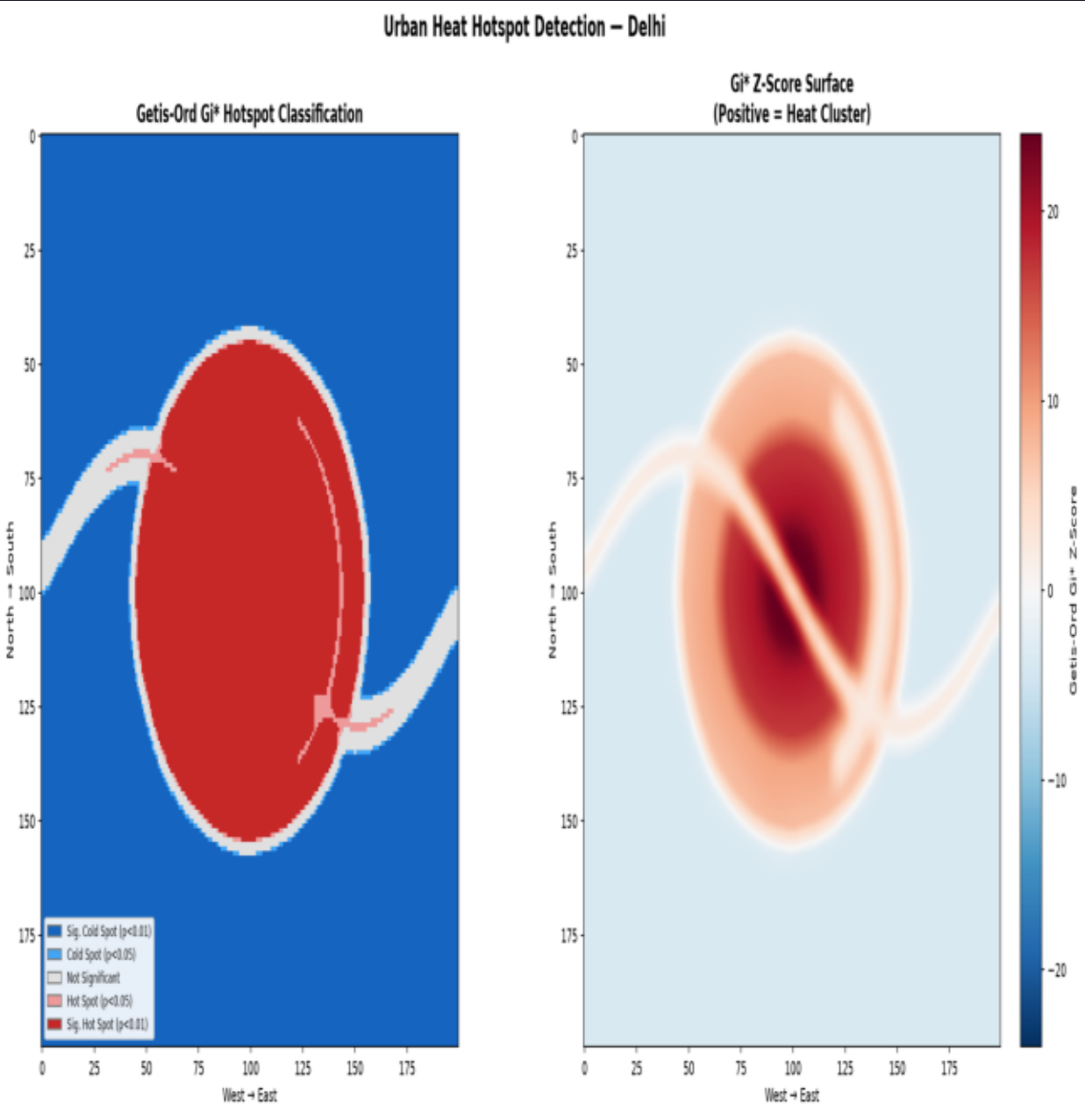
UTCI Heat Stress (6 Levels)



Urban Heat Hotspot Detection — Getis-Ord Gi*

City: Delhi

Gi* Classification + Z-Score Surface



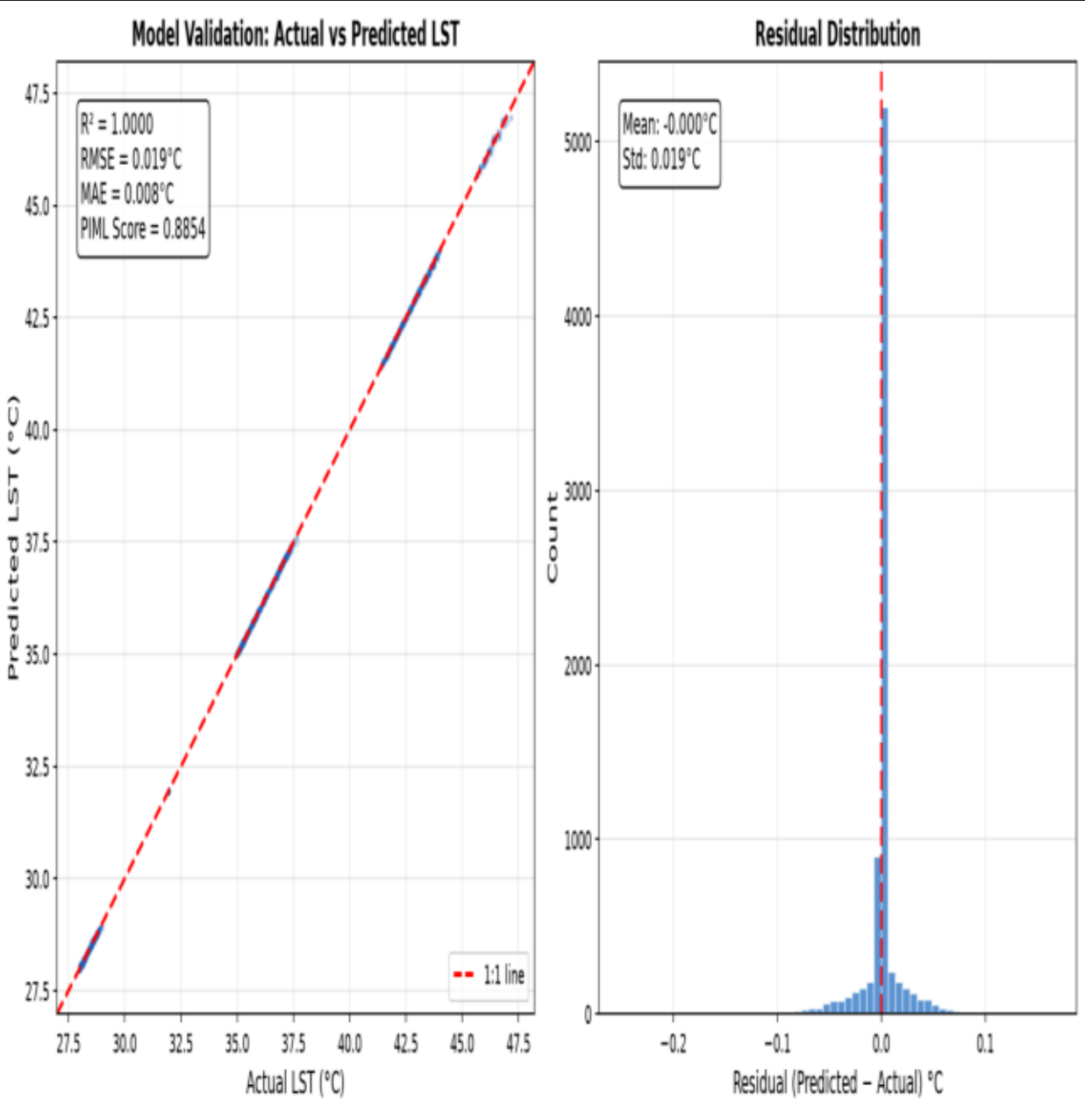
Hotspot Statistics

Total Cells:	40,000
Hotspot Cells:	9,896
Hotspot Fraction:	24.74%
Extreme Hotspots:	9,537
Cold Spot Cells:	28,424
Mean LST (Hotspot):	37.89°C
Mean LST (City):	30.52°C
Heat Excess:	+7.37°C
Max Gi* Z-score:	23.870

Physics-Informed ML Model — Validation

XGBoost + Energy Balance Constraint | R²=1.0000

Validation: Actual vs Predicted LST



Model Metrics

R² Score:	1.0000
RMSE:	0.019°C
MAE:	0.008°C
Physics Penalty:	0.7515
PIML Score:	0.8873
Features:	16
Train Samples:	32,000
LSTM R²:	0.9450
LSTM RMSE:	1.510°C

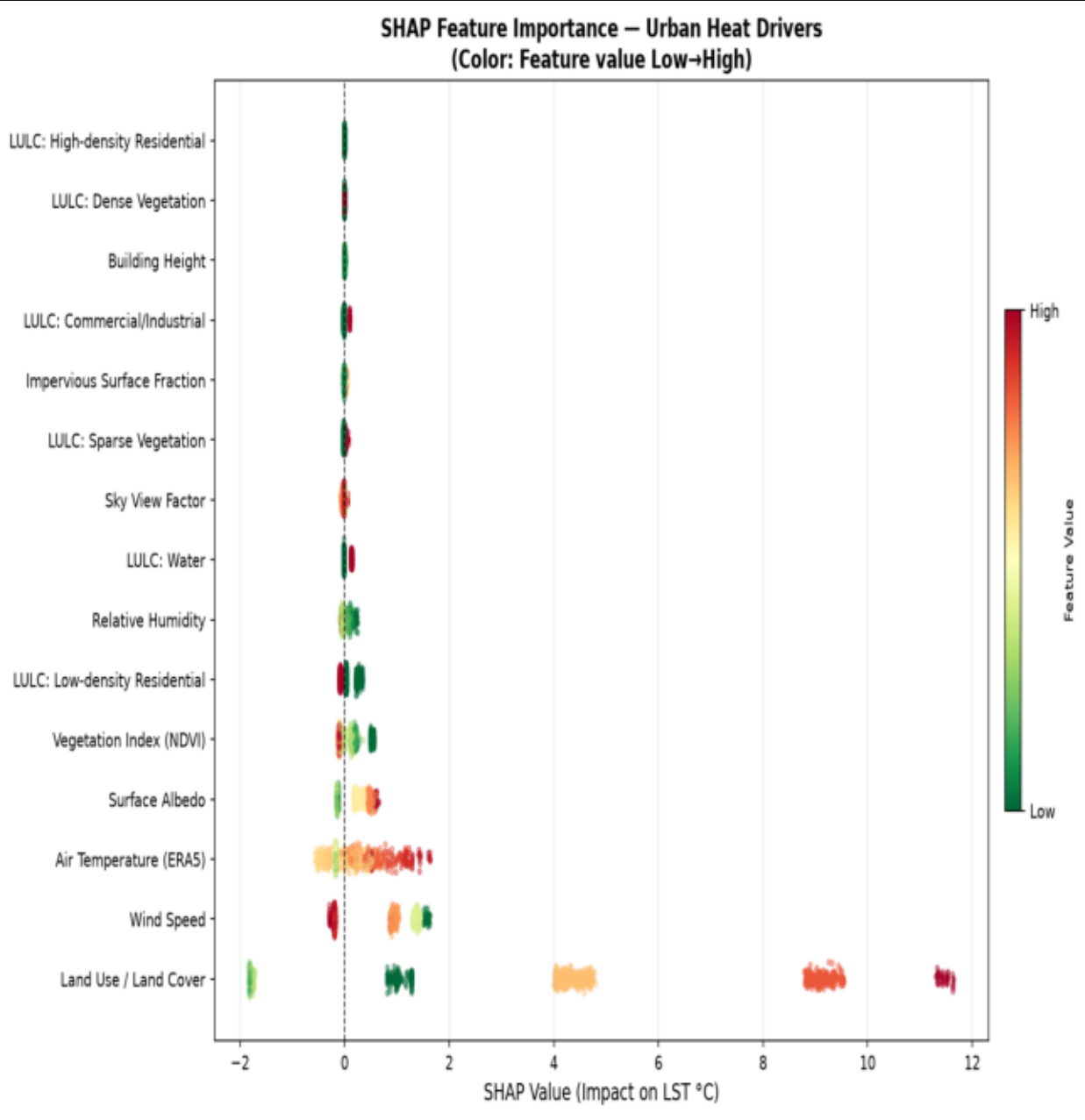
Physics Constraint

$$R_n = H + LE + G$$
$$\text{Penalty} = \lambda \times |R_n - H - LE - G|^2$$
$$\lambda = 0.15$$

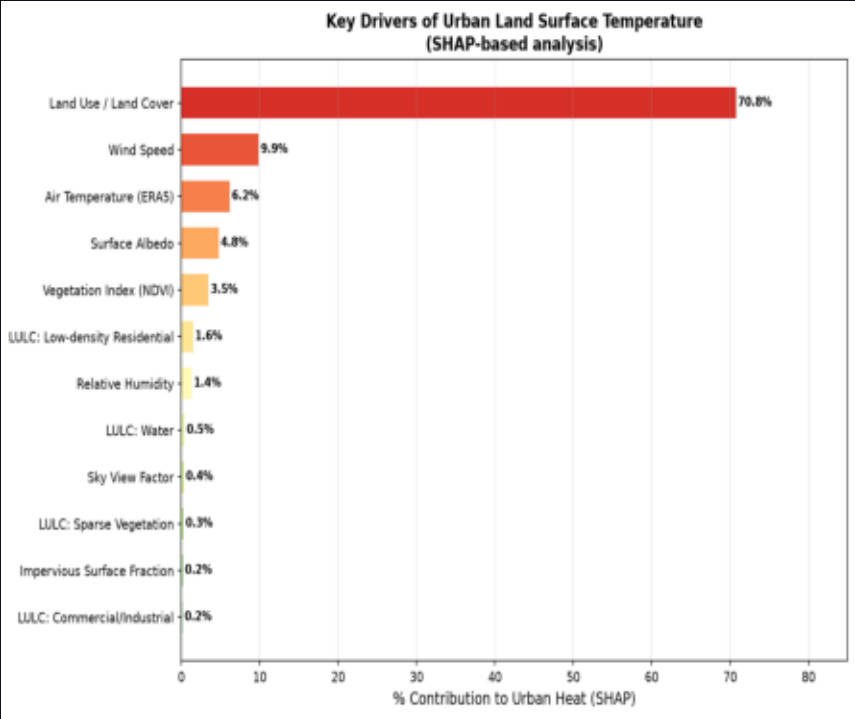
Urban Heat Driver Analysis — SHAP

SHapley Additive exPlanations | Feature Importance Decomposition

SHAP Beeswarm Plot



Top Drivers (% Contribution)



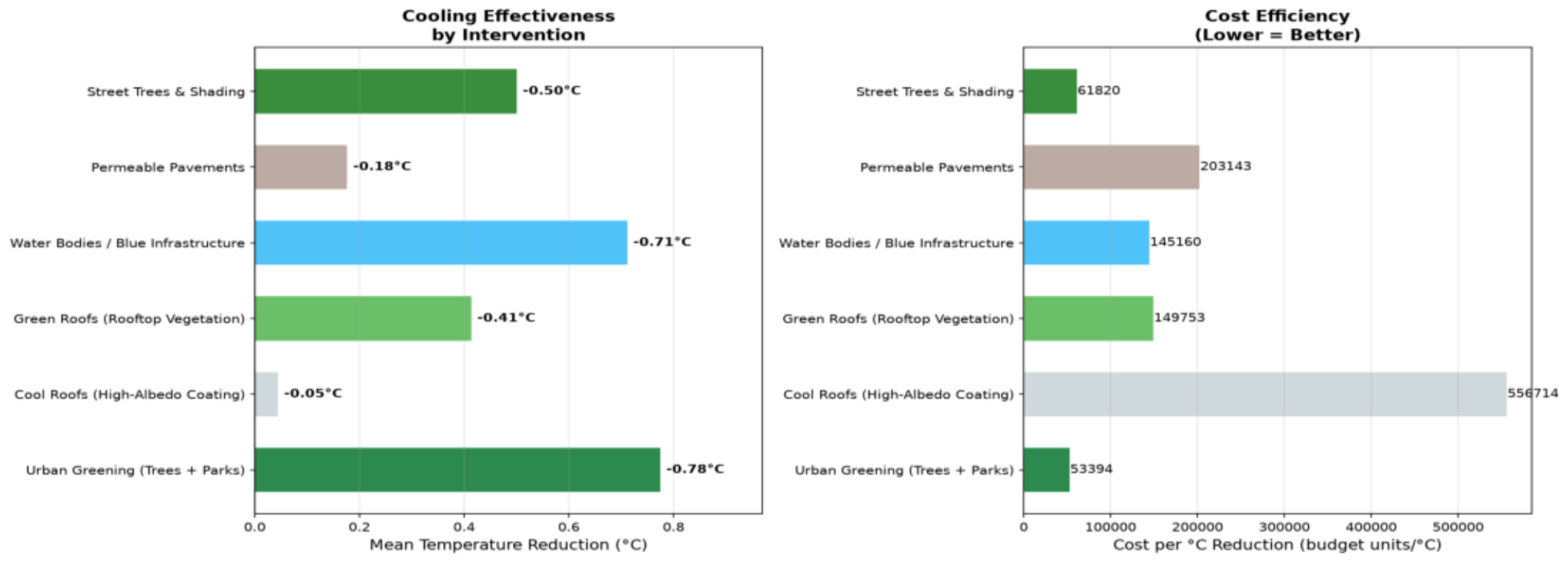
Driver Rankings

- | | |
|----------------------------|-------|
| 1. Land Use / Land Cover | 70.8% |
| 2. Wind Speed | 9.9% |
| 3. Air Temperature (ERA5) | 6.2% |
| 4. Surface Albedo | 4.8% |
| 5. Vegetation Index (NDVI) | 3.5% |

6 Intervention Types | 51.8 km² Coverage Each

Cooling Effectiveness vs Cost Efficiency

Cooling Scenario Comparison — All Interventions

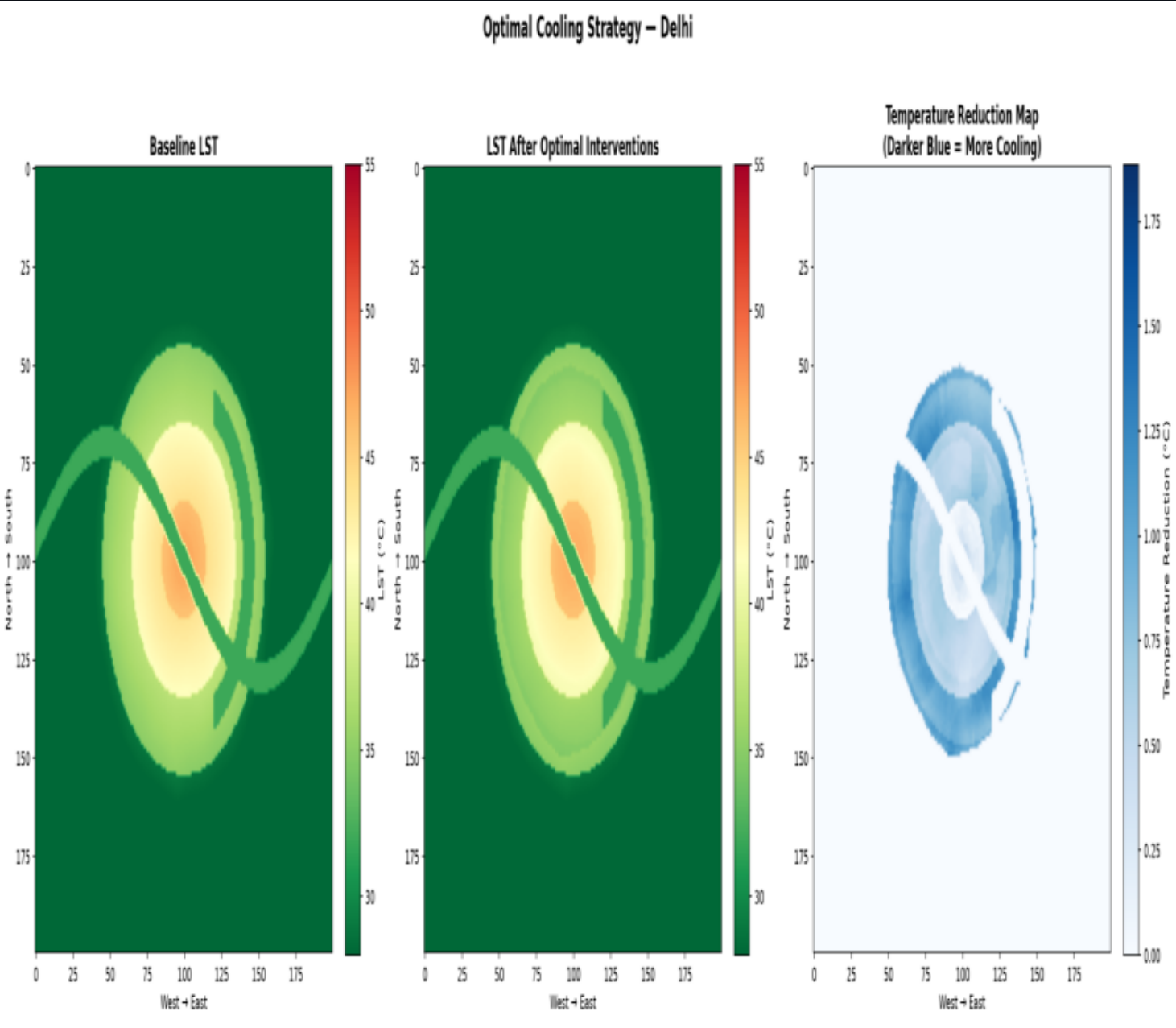


Intervention	Area (km²)	Mean ΔT (°C)	Max ΔT (°C)	Cost/°C
Urban Greening (Trees + Parks)	51.77	-0.776	-1.386	53394.3
Water Bodies / Blue Infrastructure	51.77	-0.713	-1.264	145160.2
Street Trees & Shading	51.77	-0.502	-0.912	61819.9
Green Roofs (Rooftop Vegetation)	51.77	-0.415	-0.95	149753.2
Permeable Pavements	51.77	-0.178	-0.559	203142.9
Cool Roofs (High-Albedo Coating)	51.77	-0.046	-0.27	556713.8

Optimal Cooling Strategy

Budget: 50,000 units | Area: 62.5 km² | Cooling: -0.47°C

Baseline LST | Post-Intervention LST | Temperature Reduction (ΔT)



Recommendation

Budget Used:	50,000 units
Budget Total:	50,000 units
Area Covered:	62.5 km²
Cells Treated:	6,250
Hot. Cooling:	-0.469°C
City Cooling:	-0.116°C
Intervention Mix:	
• Urban Greening	6,250

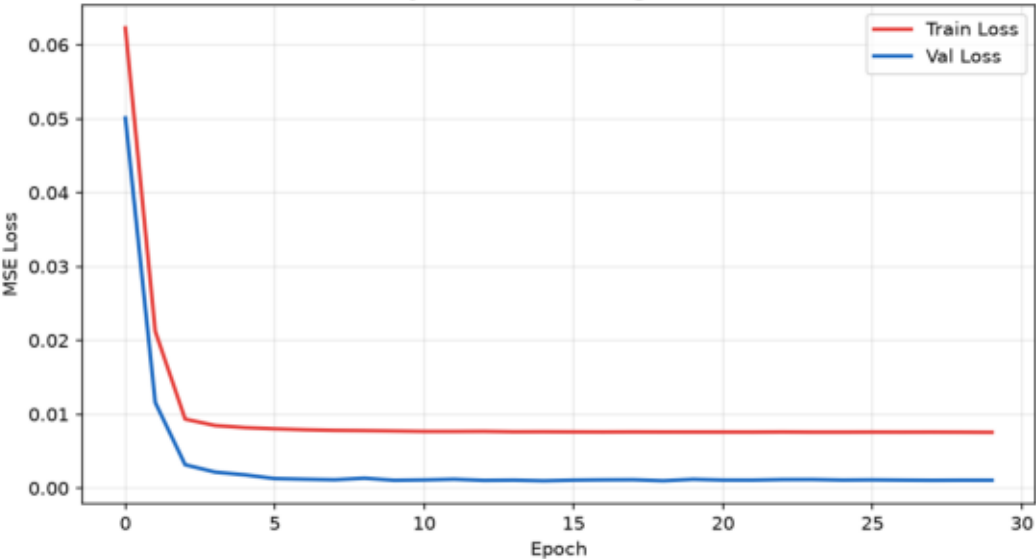
Temporal Heat Dynamics — Physics-Informed LSTM

365 days × 24h | $R^2=0.9450$ | $RMSE=1.510^{\circ}C$

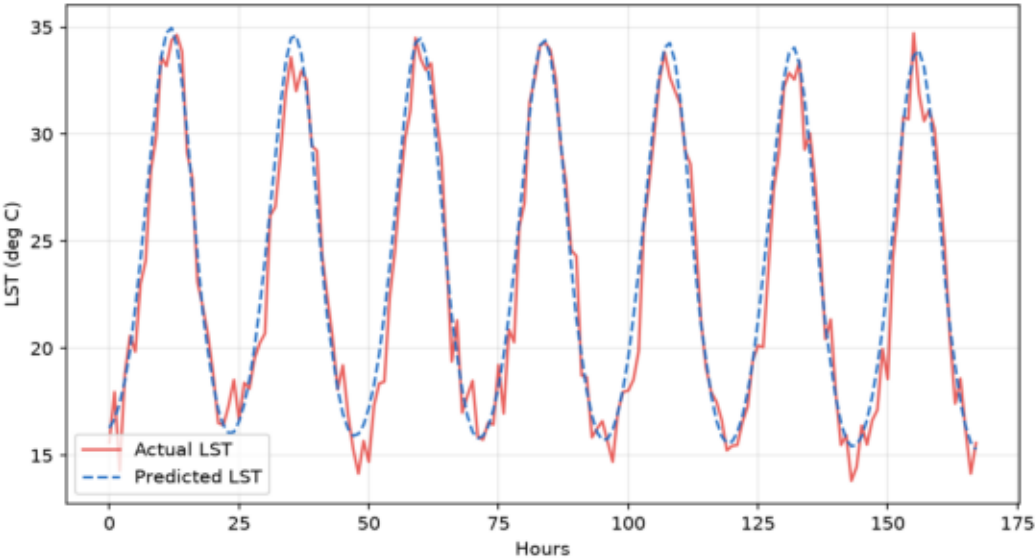
Training Curves | 7-Day Prediction | Scatter | Diurnal LULC Profiles

Physics-Informed LSTM — Urban Heat Temporal Dynamics

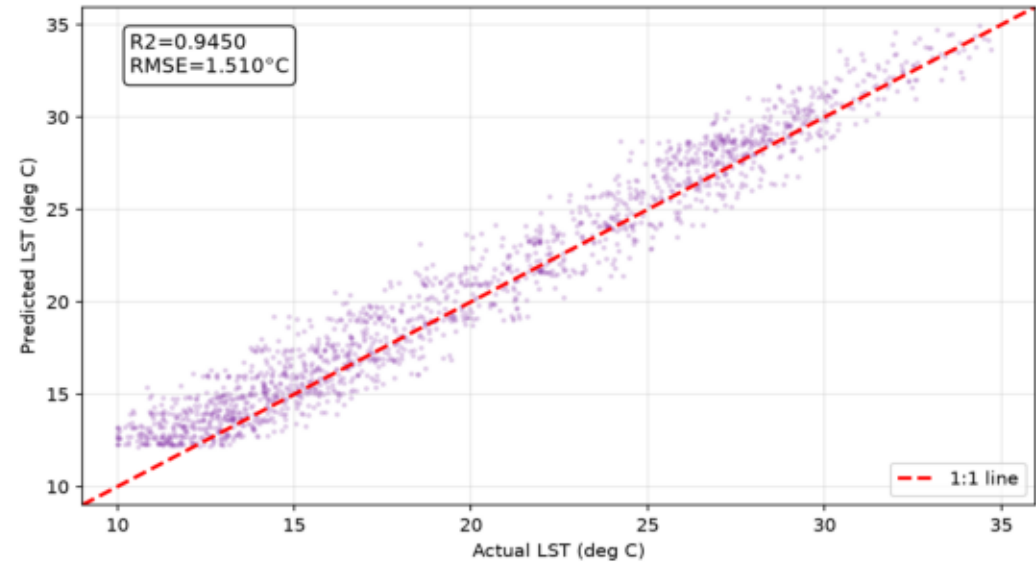
PhysicsLSTM Training Curves



7-Day Temporal Prediction
 $R^2=0.9450$ $RMSE=1.51^{\circ}C$



Actual vs Predicted LST



Diurnal LST Profile by Land Use

